



ALLIANCE
UNIVERSITY

NAAC
GRADE A+
ACCREDITED UNIVERSITY

DATABASE MANAGEMENT SYSTEM
DBMS ASSIGNMENT

ONLINE SHOPPING MANAGEMENT SYSTEM

A Relational Database Design for Managing Online Shopping Operations

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Course	BCA – Artificial Intelligence & Machine Learning
Subject	Database Management System
University	Alliance University
Academic Year	2026–2027

OBJECTIVE

To design a complete Entity Relationship (ER) Diagram for the Online Shopping Management System and convert it into a Relational Schema by identifying entities, attributes, primary keys, foreign keys, relationships, and cardinalities.

ENTITIES AND ATTRIBUTES

Entity 1: CUSTOMER

Column	Key
customer_id	Primary Key
customer_name	-
email	-
phone	-
address	-

Entity 2: CATEGORY

Column	Key
category_id	Primary Key
category_name	-
description	-

Entity 3: PRODUCT

Column	Key
product_id	Primary Key
product_name	-
price	-
stock_quantity	-
category_id	Foreign Key

supplier_id	Foreign Key
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Entity 4: SUPPLIER

Column	Key
supplier_id	Primary Key
supplier_name	-
email	-
phone	-

Entity 5: ORDERS

Column	Key
order_id	Primary Key
customer_id	Foreign Key
order_date	-
total_amount	-

Entity 6: ORDER_ITEM

Column	Key
order_item_id	Primary Key
order_id	Foreign Key
product_id	Foreign Key
quantity	-
unit_price	-

Entity 7: PAYMENT

Column	Key
payment_id	Primary Key
order_id	Foreign Key
payment_date	-
payment_method	-
payment_status	-

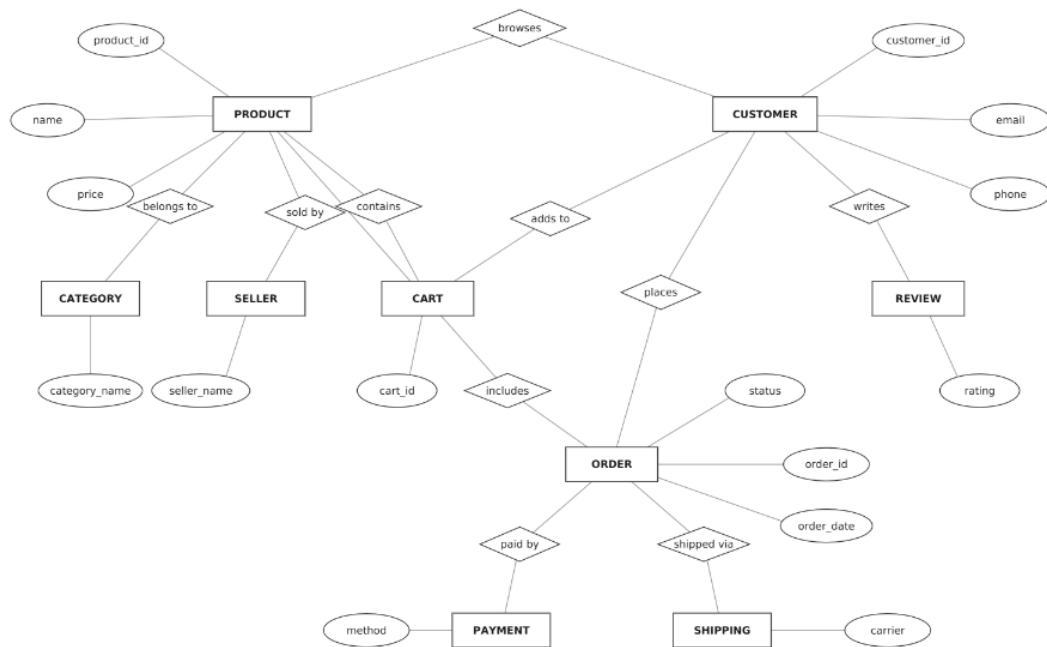
Entity 8: DELIVERY

Column	Key
delivery_id	Primary Key
order_id	Foreign Key
delivery_date	-
delivery_address	-
delivery_status	-

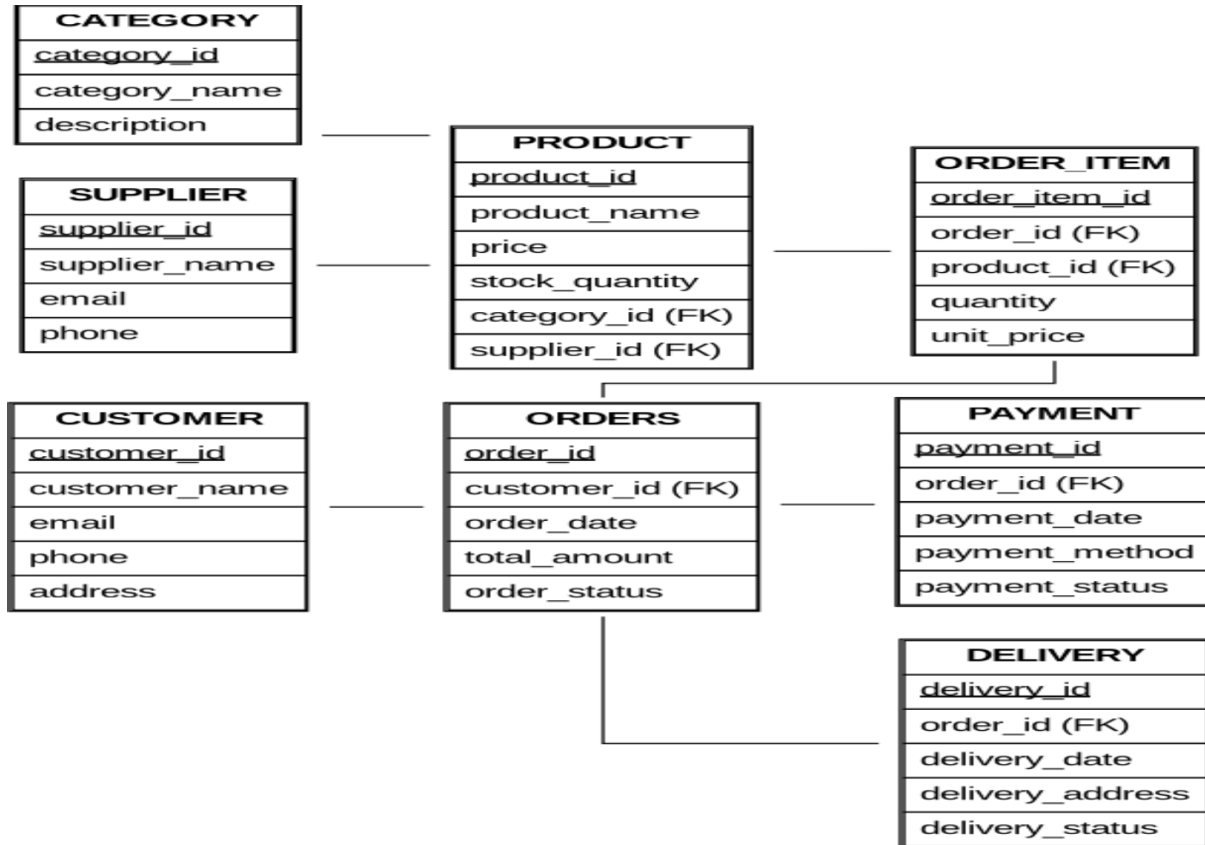
RELATIONSHIPS AND CARDINALITIES

Entity 1	Relationship	Entity 2	Cardinality
CUSTOMER	Places	ORDERS	1:M
CATEGORY	Categorizes	PRODUCT	1:M
SUPPLIER	Supplies	PRODUCT	1:M
ORDERS	Contains	ORDER_ITEM	1:M
PRODUCT	Appears in	ORDER_ITEM	1:M
ORDERS	Has	PAYMENT	1:1
ORDERS	Assigned to	DELIVERY	1:1

ER DIAGRAM



RELATIONAL SCHEMA



PRIMARY KEYS AND FOREIGN KEYS

Table	Primary Key	Foreign Key
CUSTOMER	customer_id	-
CATEGORY	category_id	-
PRODUCT	product_id	category_id, supplier_id
SUPPLIER	supplier_id	-
ORDERS	order_id	customer_id
ORDER_ITEM	order_item_id	order_id, product_id
PAYMENT	payment_id	order_id

DELIVERY	delivery_id	order_id
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RELATIONSHIP EXPLANATION

Customer - Orders: One customer can place many orders; each order belongs to one customer. Therefore, the relationship is 1:M.

Category - Product: One category can contain multiple products. Therefore, the relationship is 1:M.

Supplier - Product: One supplier can supply multiple products. Therefore, the relationship is 1:M.

Orders - Order_Item: One order can contain multiple order-item records. Therefore, the relationship is 1:M.

Product - Order_Item: One product can appear in many order-item records. Therefore, the relationship is 1:M.

Orders - Payment: Each order has one payment record in this simplified model. Therefore, the relationship is 1:1.

Orders - Delivery: Each order has one delivery record in this simplified model. Therefore, the relationship is 1:1.